

# HEALTHCONNECT — WEEK 7

## Testing Evidence Appendix

### Purpose

Visual evidence supplied by the tracks for Week 7. These screenshots support the written testing records and are not treated as evidence for activities they do not visibly demonstrate.

### Supporting visual evidence

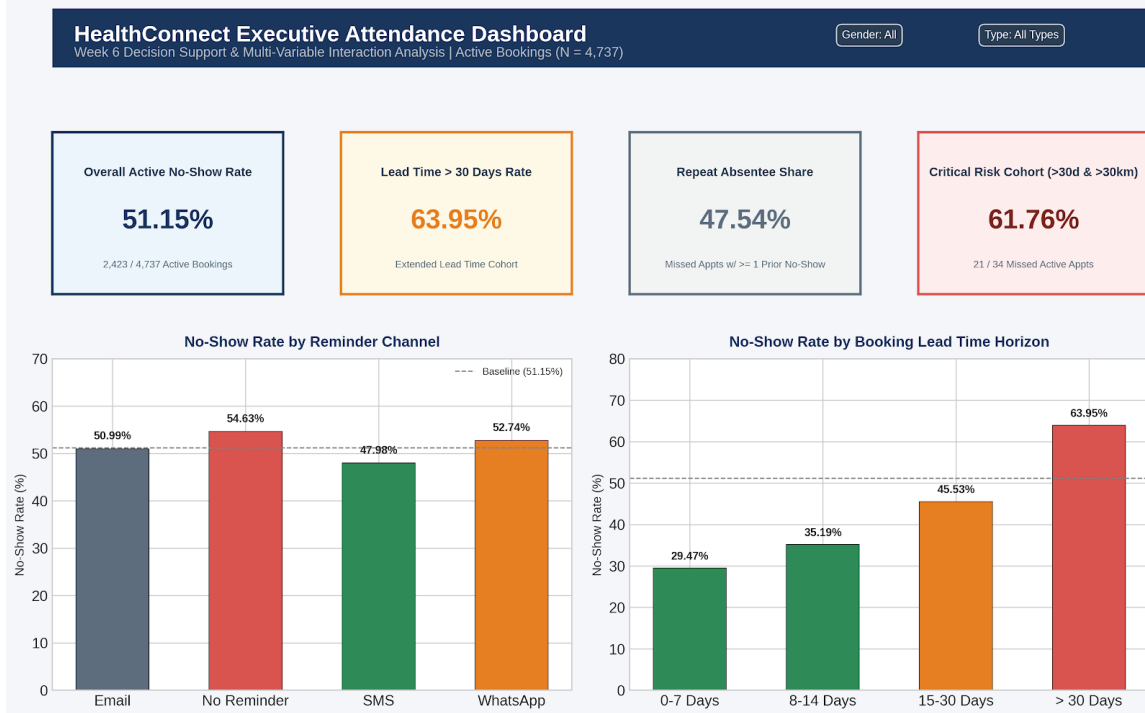


Figure 1 — HealthConnect Executive Attendance Dashboard

| HEALTHCONNECT WEEK 7 – ML PIPELINE TESTING SUMMARY  |                                   |      |
|-----------------------------------------------------|-----------------------------------|------|
| ML-01                                               | Valid input prediction            | PASS |
| ML-02                                               | Missing required field            | PASS |
| ML-03                                               | Empty / invalid numerical input   | PASS |
| ML-04                                               | Unexpected categorical values     | PASS |
| ML-05                                               | Feature engineering validation    | PASS |
| ML-06                                               | Preprocessing + model integration | PASS |
| ML-07                                               | Batch prediction                  | PASS |
| ML-08                                               | Model persistence and reload      | PASS |
| ML-09                                               | Configurable risk threshold       | PASS |
| ML-10                                               | Reproducibility and environment   | PASS |
| REF-01                                              | Input validation refinement       | PASS |
| REF-02                                              | Invalid-input retest              | PASS |
| REF-03                                              | Valid-input regression retest     | PASS |
| Tests passed: 13/13                                 |                                   |      |
| Pass rate: 100.0%                                   |                                   |      |
| PASS: All recorded Week 7 ML pipeline tests passed. |                                   |      |

Figure 2 — Week 7 ML Pipeline Testing Summary

```

INFO | Received prediction request for 1 record(s).
ERROR | Input validation failed: Numeric range validation failed: 'previous_no_shows' has 1 value(s) outside [0, 10000]
INFO | Received prediction request for 1 record(s).
INFO | Prediction succeeded. mean_probability=0.4434, n_flagged=0

negative_age:
  PASS - rejected with: Numeric range validation failed: 'age' has 1 value(s) outside [0, 120]

negative_booking_lead_days:
  PASS - rejected with: Numeric range validation failed: 'booking_lead_days' has 1 value(s) outside [0, 365]

negative_distance_to_clinic_km:
  PASS - rejected with: Numeric range validation failed: 'distance_to_clinic_km' has 1 value(s) outside [0, 500]

negative_previous_no_shows:
  PASS - rejected with: Numeric range validation failed: 'previous_no_shows' has 1 value(s) outside [0, 10000]

Valid record regression check:
  Prediction: 0
  Probability: 0.4434
  Valid prediction: True

=====
PASS: All invalid-input scenarios were rejected correctly.
PASS: Valid-input prediction still works after refinement.
PASS: Week 7 input validation refinement passed retest.

```

*Figure 3 — ML Invalid-Input Validation and Regression Retest*

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ML-02 - MISSING REQUIRED FIELD TEST
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Dropped column: 'distance_to_clinic_km'
Remaining columns: ['gender', 'age', 'age_group', 'appointment_type', 'appointment_day', 'appointment_time', 'booking_lead_days', 'previous_appointments', 'previous_no_shows', 'reminder_sent', 'reminder_channel']

Exception raised: True
Error detail: KeyError: 'distance_to_clinic_km'

PASS: ML-02 - pipeline correctly rejected an incomplete input.

```

*Figure 4 — ML Missing Required Field Test*

HealthConnect Clinic — Executive Analytics & Testing Dashboard



Figure 5 — HealthConnect Clinic Executive Analytics & Testing Dashboard